

CRUX

Game Design Document

v0.1 — Concept Phase

Real-World Scanned MMO for PC

1. Concept Overview

Crux is a PC-first massively multiplayer online action game set in a real-world environment scanned by drone-mounted LIDAR technology. Players take the role of the Crux — a species of sentient humanoids approximately one inch tall who have coexisted alongside humanity for as long as humans have existed, living undetected in the gaps, margins, and infrastructure of human cities.

The game world is built from photogrammetric scans of real locations. A city plaza, a neighborhood block, a university campus — each becomes a vast explorable zone at Crux scale, where a park bench is an overpass and a storm drain is a dungeon entrance. Maps are released incrementally and purchased by players, with the broader community able to submit their own scanned regions and earn commission on sales.

At its core, Crux is a physics-driven PVP MMO with a living AI economy, faction politics, and an urban cyberpunk aesthetic rooted in necessity rather than style — a civilization that learned to build small, build dense, and build smart because it had no other choice.

2. Setting & Lore

2.1 The Crux Species

The Crux are not a fantasy race or a science fiction invention. They are a parallel species — as old as humanity, as culturally complex, and entirely unacknowledged by the civilization that towers above them. They did not shrink. They were never human. They have always been this size, and they have built their world accordingly.

At approximately one inch tall (roughly 25mm), the Crux experience human cities as a geological landscape. Human activity is not malicious — it is weather. A construction project is a tectonic event. A dropped coffee cup is a building material. A crack in pavement is a canyon road. Generations of Crux civilization have developed entirely within the margins of human urban sprawl, invisible by proximity rather than by magic.

2.2 Aesthetic & Tone

The visual and cultural tone of Crux is cyberpunk in the genre sense — not the neon-soaked corporate dystopia of popular media, but the original literary tradition: technology outpacing social equality, dense layered urbanism, infrastructure as class divide, and advanced engineering born from constraint. The Crux build upward in pavement cracks. They string power lines between drain grates. They run transit through spaces no human has looked at twice.

This aesthetic emerges naturally from their environment rather than being imposed upon it. The result is a civilization that looks genuinely alien until you recognize the human objects repurposed at a different scale — and then it looks inevitable.

2.3 Relationship With the Human World

Humans are not antagonists. They are an environmental force — vast, indifferent, and occasionally catastrophic. A pigeon landing nearby is a natural disaster. A human child crouching to look at something on the ground is a first-contact event. This dynamic creates tension without requiring a villain, and it grounds the fiction in something players can immediately relate to from the opposite perspective.

3. World & Map System

3.1 LIDAR Scanning Pipeline

Game maps are generated from real-world drone LIDAR scans. The scanning workflow is:

- FPV drone with mounted LIDAR sensor flies a planned route over a target area
- Raw point cloud data is post-processed for GPS drift and coverage gaps
- ML classification pipeline identifies surface types (concrete, grass, metal, wood, water, etc.)
- Game-ready mesh is generated, LOD-optimized via Unreal Engine 5 Nanite, and navmesh-baked
- Human QA review validates surface classification and patches mesh holes before release

Unreal Engine 5's Nanite system handles photogrammetry mesh complexity natively, eliminating the need for aggressive manual decimation on PC. Mobile optimization (aggressive polygon reduction with normal map baking) is a future phase consideration.

3.2 Scale

Player characters are one inch (25mm) tall, equivalent to approximately 1:72 real-world scale. At this scale:

Real Distance	In-Game Distance	Notes
1 foot	72 feet	Standard room = city block
100 feet (plaza)	~1.4 miles	Single plaza = full MMO zone
660 feet (city block)	~9 miles	One block = continent-sized content

Real Distance	In-Game Distance	Notes
Real ant (~1.5mm)	~1/17th player height	Fast, aggressive, manageable
Real spider (~25mm)	Player-sized	Peer-level threat, genuinely terrifying
Real pigeon	~10-story building	Environmental hazard

This scale means a small number of real-world scanned locations generates an enormous amount of playable content, making the scanning investment highly efficient per hour of gameplay.

3.3 Zone Structure

The game world is divided into zones corresponding to distinct real-world locations. Zones are categorized by their real-world character:

Zone Type	Real-World Equivalent	Game Function
Safe	Town squares, plazas, parks	Trading, loadout swapping, social hubs
Contested	Streets, parking lots, open paths	PVP likely but not guaranteed
Dangerous	Alleys, construction sites, dense cover	High risk, high reward

Zone danger classification is determined automatically during the ML pipeline pass based on cover density and sight line complexity, then validated by human QA.

3.4 Seamless Zone Transitions

Adjacent zones run on separate dedicated server instances. Transitions between zones are masked by real-world geography — a road, a wall, a fence provides a natural visual boundary behind which the player is silently transferred to the neighboring server instance. Both zone servers run simultaneously and sync player state at boundaries, making the transition imperceptible.

This architecture is inspired by techniques used in games with seamlessly instanced spaces (vehicle interiors, building transitions) and is well-suited to Crux because real geography provides clean, natural handoff points unlike arbitrary invisible lines.

3.5 Map Monetization & Community Submissions

The game client is free. Maps are the purchasable content. The starter zone (a city plaza) is available at no cost and is designed to be generous enough to establish the hook before any paywall is encountered.

Additional map regions are unlocked individually at a price point appropriate to their size and content density. Community members may submit their own LIDAR scans through a formal submission pipeline. Accepted submissions earn the submitter a commission (approximately 30-40%) on every unlock sale for that region. Alternatively, the studio may offer a flat-fee buyout for high-traffic locations.

Community submissions must pass the same ML classification and human QA pipeline as official scans. Automated seam validation checks that submitted scans align correctly with adjacent zones in mesh scale and surface classification before approval.

4. Core Gameplay

4.1 Design Philosophy

Crux is a physics-first game. Interactions emerge from systems rather than scripted events. Real-world geometry at 1-inch scale is inherently unpredictable and interesting — a storm drain grate is a dungeon entrance, a park bench is a highway overpass — and the gameplay leans into this rather than smoothing it away. Players discover interactions the designers never explicitly planned, which extends content longevity and creates genuine emergent storytelling.

4.2 Movement & Combat

Combat feel draws from dash-based action games with committed momentum. The dash mechanic is inspired by Pyra/Mythra's side special in Super Smash Bros. Ultimate, adapted without invincibility frames:

- Wind-up telegraph: a short animation before the dash fires gives opponents a reaction window
- Committed momentum: players are locked into their dash direction with no mid-travel correction
- Momentum carry: dashing off ledges converts to airborne momentum, creating terrain-dependent plays
- End-lag punish window: missed or wall-colliding dashes result in a brief stagger

The absence of iframes means dashes are a repositioning and spacing tool rather than a panic button. Players who dash preemptively — reading opponent telegraphs — are rewarded. Dash-mashing is naturally punished by end-lag without requiring explicit counters to be designed.

Physics interaction with the environment is a core combat layer. Scanned objects at 1-inch scale (debris, litter, small urban objects) can be pushed, thrown, and used as improvised cover. Vertical play is significant — high ground, ledge ambushes, and fall mechanics all carry real weight at this scale.

4.3 Loadout System

Each player has a fixed number of loadout slots: weapon, tool, consumable, and traversal. What fills those slots determines playstyle. Progression is horizontal — new unlocks are different, not stronger. A veteran's unlocked grapple hook is not better than a new player's dash boots; they create different playstyles with different counters.

The starter loadout is intentionally complete and genuinely viable. New players are not missing something critical — they lose to veterans on execution and game knowledge, not on stat gaps. This distinction is critical to retaining casual players while still providing meaningful progression for dedicated ones.

4.4 Death & Economy

Death causes the player to drop a portion of their carried resources (cash and crafting materials). Unlocked loadout pieces are never dropped on death. Veterans carry more resources and therefore have more to lose from dying in contested territory, which creates natural risk-reward scaling without touching base stats or power levels.

5. Faction System

5.1 Overview

The faction system is modeled on Sea of Thieves' trading company structure. Factions do not lock players out of content — they provide a lens through which to engage with it. Players may pledge to multiple factions simultaneously, though reputation gains are split, incentivizing eventual specialization without forcing it.

Faction identity comes from how players fight, not where. All factions compete for the same objectives in the same zones. The game avoids geographic faction segregation by ensuring contested objectives spawn across all terrain types on timers, pulling all factions to the same point regardless of their playstyle preferences.

5.2 Faction Roster

Faction	PVP Style	Perk Direction	Dash Variant
Ravagers	Aggressive, direct combat	Damage, armor, sustain	Short, explosive, ends in a hit
Phantoms	Ambush, stealth, traps	Burst, invisibility, escape	Leaves a decoy at origin point
Wardens	Defensive, territory holders	Shields, area denial, fortification	Plants a temporary shield at destination
Brokers	Economic PVP, bounties	Intel, tracking, resource denial	Standard, but marks opponent for allies
Drifters	Mobility, hit-and-run	Enhanced traversal, verticality	Longest range, most momentum, worst end-lag

Each faction's dash variant uses the same core mechanic with different strategic expression. No dash is objectively superior — they counter each other based on terrain and matchup, creating faction-level rock-paper-scissors dynamics without vertical power gaps.

The Broker faction functions as economic PVP — placing bounties, manipulating markets, hiring other factions for work. They are the intelligence and financial layer of the faction ecosystem and do not need to win fights directly to be effective.

5.3 Objective Structure

Contested objectives spawn across all zone types on timers, inspired by Sea of Thieves' event system but driven by NPC behavior rather than map markers. Players learn objective locations through observation and player-driven information sharing rather than UI notifications — knowing where a high-value NPC caravan is right now is itself a tradeable commodity.

6. Economy & NPC Systems

6.1 Currency

The in-game currency is cash — recognizable, universally understood, requiring no tutorial. Bills at Crux scale are physical objects in the game world. Economic complexity is expressed through NPC and player behavior rather than currency mechanics, keeping the system accessible to casual players.

6.2 AI Corporation Owners

The game economy is driven by several autonomous AI corporation entities, each running on a small locally-hosted LLM (target: Mistral 7B or equivalent via Ollama). Each corporation reasons

about its own economic state and makes decisions on a slow tick — every few in-game hours — rather than in real time. This keeps inference costs manageable while producing meaningful emergent behavior.

Each corporation tracks and reasons about: supply and demand across its trade routes, faction reputation and relationships, competitor pricing, player behavior history, and route safety. Decisions are discrete (reroute, raise price, hire escorts, issue bounty, form alliance) rather than freeform, keeping the LLM context window small and inference fast.

Corporations develop identity over time through their history. A corporation repeatedly raided by Ravagers becomes hostile to that faction, fortifies its shops, and reroutes its caravans. A thriving corporation expands — claiming new shop locations, hiring more NPC carriers, developing a recognizable brand that players learn. Bankrupted corporations disappear, creating resource shortages players must respond to.

6.3 Physical Shops & Trade Routes

Corporation shops are physical objects in the scanned world — soda cans, matchboxes, bottle caps repurposed as storefronts. These become landmarks players navigate by. Corporation NPC caravans physically travel routes between shops, crossing safe zones, contested streets, and dangerous alleys on schedules players can learn and exploit.

Shop ownership operates on three tiers:

- **Corp:** Corporation-owned: AI-stocked and AI-priced, reliable but impersonal. AI may close shops in dangerous areas if losses are too high.
- **Player:** Player-owned: Players claim physical objects as shop fronts, stock from their own inventory, set their own prices. Direct competition with AI pricing.
- **Contested:** Contested: High-value locations that multiple corporations and players want. Ownership requires active defense, functioning as a persistent PVP driver.

6.4 Player-Built Transportation

Transportation infrastructure is player-built and faction-operated rather than designer-placed. The Crux civilization has always been small and has engineered accordingly — transportation reflects this, with dense, efficient systems built in the margins of human infrastructure.

Infrastructure scales with the game world as new map regions are released:

- Local scale (few blocks): creature mounts, foot travel
- City scale (multiple districts): player-built transit lines operated by factions or corporations
- Regional scale (multiple cities): larger engineered transport connecting scanned areas
- Continental scale (future): high-speed infrastructure added when world size justifies it

AI corporations respond economically to player transportation. A player-built rail line cutting into a corporation's foot traffic is a competitive threat the AI entity will react to — rerouting, undercutting prices, or hiring factions to disrupt the competing line.

7. Technical Architecture

7.1 Engine

Unreal Engine 5 (C++ / Blueprints). Selected for Nanite (photogrammetry mesh handling), Chaos physics (TotK-style object interaction), Lumen (dynamic global illumination on scanned geometry), and mature dedicated server networking.

7.2 Server Architecture

Layer	Technology	Purpose
Zone servers	Unreal dedicated server on AWS GameLift	One instance per active map zone, auto-scaling
Master server	Go or Rust REST API	Auth, player state, inventory, faction rep — source of truth
Database	PostgreSQL + Redis	Postgres for persistence, Redis for live session data
LLM economy	Ollama + Mistral 7B (self-hosted)	Corporation AI reasoning, decoupled from game servers
Map CDN	AWS S3 + CloudFront	Scanned geometry streamed to clients via Unreal level streaming
Submission pipeline	Isolated processing server	Community scan ingestion, ML classification, QA — never touches live game

7.3 Zone Boundary Synchronization

Zone boundary synchronization is the most complex networking challenge in the architecture. The solution is a buffer zone at boundaries — a short region where the player is tracked by both the outgoing and incoming zone server simultaneously, with the master server arbitrating state. Real-world geography provides clean handoff points (roads, walls, fences) that make this manageable compared to arbitrary invisible boundaries.

7.4 Scan Processing Pipeline

Map processing is entirely decoupled from live game infrastructure. Scans enter a dedicated pipeline server and never interact with live game systems until they have cleared ML classification, human QA, and seam validation. Processed maps are pushed to the CDN and made available to zone servers for streaming.

The ML surface classification pipeline uses 3D semantic segmentation (target architecture: RandLA-Net or PointNet++). LIDAR intensity returns and co-registered RGB camera data are used in combination for classification accuracy. The pipeline outputs surface type tags (concrete, grass, metal, wood, water, etc.) plus a derived danger tier based on cover density and sight line complexity.

7.5 Privacy & Legal

Scans are post-processed for privacy compliance before any human review. Automated detection flags human-shaped point clusters and license plate geometry for removal. A contracted 3D modeler handles clean excision and mesh patching of flagged clusters. This approach removes people from the game world entirely rather than blurring them, which is both legally cleaner and better for player immersion.

Scanning and commercializing public spaces is generally permissible. Private property, recognizable storefronts, and international scans (GDPR/CCPA) are reviewed on a case-by-case basis before approval.

8. Business Model

8.1 Revenue Structure

- Free client download with a fully playable starter zone (city plaza)
- Additional map regions sold individually at pricing proportional to size and content density
- No battle passes, no cosmetic MTX at launch — maps are the content and the monetization
- Community submission commissions (~30-40% of unlock revenue) paid to approved submitters

8.2 Platform

PC (Steam) at launch. Mobile expansion is a future phase contingent on polygon reduction pipeline development and market validation. Console is not planned in the near term.

8.3 MMO Structure

Crux is a persistent online world. The free client is non-functional without at least one purchased map region, creating a clear and honest paywall that players understand before downloading. The game functions as a traditional MMO within purchased regions — persistent world, multiple simultaneous players, player economy, faction politics.

Singleplayer or story mode is a potential future addition if player interest warrants it, but is not part of the initial design scope.

9. Team & Development Roadmap

9.1 Core Team Requirements

Role	Priority	Notes
ML / 3D Segmentation Engineer	Hire First	Must have specific 3D semantic segmentation experience (RandLA-Net, PointNet++). Defines pipeline speed and quality.
Scanning / Drone Operator	Hire Second	LIDAR data quality expertise — flight path planning, overlap ratios, GPS drift correction, point cloud post-processing. Can be part-time contractor early.
3D Modeler (Privacy)	Hire with Operator	Human and license plate removal from scans. Mesh patching. Freelance per-scan initially.
Map QA Reviewer	Hire Third	Reviews ML classification output, fixes surface misclassification, verifies navmesh. Remote freelance per map initially.

9.2 Development Phases

Phase	Goal	Key Milestone
1 — Proof of Concept	Scan starter plaza, import into Unreal 5, walk around it	Playable scan exists
2 — Pipeline	Automate scan-to-mesh-to-classify-to-game workflow	ML Engineer hired, pipeline functional
3 — Alpha	One playable zone, basic MMO skeleton, faction system, closed testers	10-50 internal testers
4 — Monetization	Free client launch, plaza + 2-3 purchasable zones, gauge demand	First revenue
5 — Economy	LLM corporation AI deployed, NPC caravan routes live	Living economy running

Phase	Goal	Key Milestone
6 — Community	Submission pipeline open, community scanning program launched	External map contributions
7 — Scale	Transportation infrastructure, request system, city expansion	Multi-city world

10. Open Questions

The following design areas are identified but not yet resolved:

- Core PVP objective structure — resource nodes, territory control, or bounty system as the primary driver pulling factions into conflict
- Exact faction perk trees — horizontal progression specifics for each faction beyond dash variants
- Player housing / base building — whether players can establish persistent structures in the game world
- Creature system — which real-world insects and animals appear, at what threat tier, and how they interact with faction and economy systems
- Trademark clearance — Crux name search through USPTO before filing
- Early access strategy — whether Phase 3 alpha becomes a paid early access release or remains closed